Trevor Erwin

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Education

August 2024 - Present

Purdue University - *Ph.D. in Planetary and Atmospheric Sciences*

September 2020 - December 2023

University of Texas at Austin- *Honors Bachelors of Science in Astronomy and Physics*

Graduated in December of 2023 with a GPA of 3.75.

Awards

NSF Graduate Research Fellowship Program Honorable Mention **2025**

- Received Honorable Mention for my proposal on Salt Clouds in Y-Dwarf Atmospheres

Purdue Presidential Excellence Ph.D. Award, **2024**

- Award presented in recognition of my research and alignment to the goals of Purdue University.

Karl G. Henize Endowed Scholarship, **2023**

- Award given based on my overall performance in regards to grades, research, and service to the department at UT Austin.

Melvin J. Rieger Scholarship Fund in Physics, **2023**

- Award given based on my research on the distribution of hollow spheres in exoplanet atmospheres. This research is further detailed under *Research Experience.*

Astronomy Dept. Research Funding & Undergraduate Research Grant, **2023**

- Two separate grants, with awards given based on my research on the distribution of hollow spheres in exoplanet atmospheres. This research is further detailed under *Research Experience.*

Olin E Teague National Space Research Scholarship, **2020**

- Award given based on my work designing polyethylene Martian habitats that I synthesized. In this work, I created two spherical model habitats and tested their ability to

survive on the Martian surface, using vacuum chambers to fill the inside with Earth-like pressure and the outside with Martian pressure. I then subjected them to extreme Martian temperatures, and tested their ability to withstand radiation that would be caused by cosmic rays.

Research Experience

August 2024 - PRESENT

Graduate Research advised by Dr. Alexandria Johnson at Purdue - *Salt Microphysics and Deliquescence in Exoplanet Cloud Models*

- Project Goal: *Study the effect of deliquescence and efflorescence of KCl salts in water clouds, and incorporate hygroscopicity into Community Aerosol and Radiation Model for Atmospheres (CARMA).*
- My Tasks:
    - Investigate the deliquescence point of KCl in presence of water vapor in a Y dwarf-like environment.
    - Incorporate the cloud microphysical properties into CARMA.

May 2024 - August 2024

Research Assistant for Dr. Ujjwal Raut at Southwest Research - *CO_2 enhancement of H_2O_2 in the Chaos Terrains of Europa*

- Project Goal: *Study the effect of CO_2 on the amount of peroxide created on Europa's surface ice via irradiation.*
- My Tasks:
    - Independently ran deposition and irradiation experiments in the laboratory. Operated the ultra high vacuum chamber setup described in Raut et al., 2022, *Science Advances*.
    - Created fully automated python code to calculate band areas from laboratory results, as well as JWST data.

June 2022 - March 2024

Research Assistant for Dr. Brianna Lacy and Dr. Caroline Morley at UT Austin - *Creating Models that Account for Mie Scattering in the Hazes of Exoplanet Atmospheres*

- Project Goal: *Investigating whether deviations from spherical haze particles will be detectable in JWST transit spectra.*
- My tasks:
    - Creating a module to incorporate into an existing transit spectrum code which implements the distribution of hollow spheres approach.
    - Modeling transit spectra with a variety of shape distributions, particle compositions, and other atmospheric properties to explore whether deviations from solid homogeneous particles will be detectable.

January 2021 - December 2021

Research Assistant for Dr. Michael Montgomery – *Determining Ages of Star Clusters by Creating Color Magnitude Diagrams and Comparing them to Existing Isochrones*

- Project Goal: *Use observational data from the Gaia Early Data Release 3 to create color magnitude diagrams for a number of Messier clusters. To determine the age of the cluster, these diagrams were overlaid with existing isochrones to determine the best fit.*
- My tasks:
    - Performing a literature review on white dwarf stars and their evolutionary life cycle.
    - Use the Python package Astropy to create a code to analyze globular clusters and create a color magnitude diagram based on luminosity. By comparing this with isochrones, I was able to determine the age of star clusters.

May 2020 - August 2020

Research Assistant for Dr. Ben Teolis, research scientist at Southwest Research Institute – *Analyzing the Methane Deposition Cycle on Charon*

- Project Goal: *Investigate the seasonal methane deposition cycle on Pluto's moon Charon. Images from New Horizons revealed that a majority of the red methane was deposited on the northern pole of Charon, and this work sought to find out why the methane seemed to exclusively lie in that region.*
- My Tasks
    - Using vacuum chambers and lasers to deposit methane in a sub-zero environment.
    - Completed collection of data and contributed to information presented in the report.
    - Helped contribute to new theories on chemical deposition and ionization in sub-zero conditions.
- Work contributed to the following publication:
    - U. Raut et. al., "Charon's refractory factory," *Science Advances,* vol. 8, no. 24, 2022.

Publications

Erwin, T. P., Johnson, B. C., Minton, D., & Johnson, A. V. (2026). The Length of Martian Crater Rays and their relation to Lunar Cold Spots. *Journal of Geophysical Research: Planets*, 131, e2025JE009609. https://doi.org/10.1029/2025JE009609

U. Raut, B. D. Teolis, T. A. Nordheim, S. Protopapa, B. D. Mamo, **T. P. Erwin**, I. Wong, J. A. Kammer, K. D. Retherford, R. J. Cartwright, T. Peterson, G. Villanueva, S. D. Vance (2026).

Ocean-sourced CO2 Boosts Radiolytic Oxidants at Europa's Chaos. *Nature Communications*. ***Work in Review***

Mamo BD, Raut U, Teolis BD, **Erwin T**, Cartwright RJ, Protopapa S, Retherford KD, Nordheim TA. (2025). *"Laboratory Investigation of CO_2-driven Enhancement of Radiolytic H_2O_2 on Europa and Other Icy Moons."* Planetary Science Journal. **6** (170). https://doi.org/10.3847/PSJ/ade3d8

Presentations

Poster Presentations

- *"The Chemistry of Clouds: Salt's Effect on Water Vapor in Y-Dwarf Atmospheres,"*
  - **Great Lakes Exoplanet Area Meeting (2025)**
- *"The Distribution of Hollow Spheres in Exoplanet Transit Spectra,"*
  - **American Astronomical Society's 243rd Meeting**
  - **Frank N. Bash Symposium, October 2023**
  - **Texas Area Planetary Science Meeting, Summer 2023**
- *"The Distribution of Hollow Spheres in Exoplanet Atmospheres,"*
  - **UT Undergraduate Research Forum, Spring 2023**
  - **UT Physics and Astronomy Undergraduate Research Forum, Spring 2023**

Oral Presentations

- *"Microphysical Pathways of Salt-Induced Cloud Formation in Ultracool Substellar Atmospheres,"*
  - **American Astronomical Society's Division of Planetary Sciences 2026 Meeting**
- *"The Chemistry of Clouds: Salt's Effect on Water Vapor in Y-Dwarf Atmospheres,"*
  - **American Geophysical Union's Society's 2025 Meeting**
- *"The Length of Martian Crater Rays and Their Relation to Lunar Cold Spots,"*
  - **American Geophysical Society's 2025 Meeting**
- *"The Chemistry of Clouds: Salt's Effect on Water Vapor in Y-Dwarf Atmospheres,"*
  - **American Geophysical Union's Society's 2025 Meeting**
- *"Finding Life On Other Worlds"*
  - **Schrodinger's Pint, Darwin's Piano Bar, December 2022**
- *"Exoplanet Habitability: Finding Life On Other Worlds"*
  - **Astronomy Students Association, Student Research Presentations, October 2022**

Employment

January 2026 - May 2026

Graduate Teaching Assistant for Introductory Geology

- Served as primary contact for student-teacher relations
- Help solicit learning for the intro geology class for STEM majors at Purdue University.

Spring 2023 - December 2023

Park Ranger and Outreach Specialist for McKinney Roughs Park

- Headed astronomy based outreach events for the park, as well as a host of nighttime events
- Experience with night vision equipment, giving guided talks on trails, and guest relations

September 2021 - December 2023

Physics Office Lab Demonstrator

- Setup physics demonstrations for introductory physics classes (e.g. mechanics, electricity and magnetism, thermodynamics, and waves)
- Complete scheduled maintenance of physics demonstrations
- Oversee orientation and training of new employees

September 2022 - December 2023

Undergraduate Teaching Assistant for Introductory Astronomy

- Served as primary contact for student-teacher relations
- Help solicit learning for the intro astronomy class for STEM majors at UT Austin.
- Responsible for grading assignments and hosting weekly office hours where I assist students and facilitate learning.

Observing Time

McDonald Observatory's 2.3m - 3 Nights

- Observed twelve faint galaxies and nebulae while operating the Otto Struve 2.1m telescope at McDonald Observatory
- Helped instruct my peers on safe telescope operations

Professional Skills

Programming

- Programming experience with python (e.g., for reading and writing multi-dimensional catalogs, array manipulation, plotting, and statistical analysis)

- 4+ years of experience with the Astropy package in Python and other astronomy-related python packages
- Experience with MacOS and Linux operating systems
- Experience with LaTeX for writing scientific papers

Certifications and Training

- Machine Shop Certification (Lathes, mills, etc)
- Laser Safety Training
- Cryogenic Safety Training
- 3D Printer Certification

Outreach and Campus Involvement

Spring of 2026

Colombia Astrocamp 2026

- Traveled to Colombia and taught students from a variety of schools about astronomy. Wrote lesson plans, gave oral presentations, and communicated in a bilingual environment.

Summer of 2025 and 2026

Purdue Physics Inside Out Summer Camp

- Wrote lesson plans and taught students about crater formation, scaling the solar system, and creating rockets during Purdue's annual physics summer camp for middle school students.

Fall 2021 - 2024

Astronomy Students Association - *Outreach Officer*

- Planned meetings, arrange for speakers inside and outside of the department, arrange semester camping trip, engage in outreach to the Austin and greater Austin school districts
- Organized outreach events in coordination with local schools to inspire the next generation of young scientists.
    - This includes elementary outreach events, trips to dark sky spots like Enchanted Rock State Park, and the planning of our annual visit to McDonald Observatory
- Advocate for the needs of UT Austin undergraduate students to departmental leadership.

Fall 2022 - 2024

Astronomy Mentorship Program (AMP) - *Mentor*

- Mentor junior undergraduate astronomy students on research opportunities, career goals, and schedule planning
- Coordinate with the graduate astronomy mentorship program to better both programs and their mentees.